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STELLAR STRATIGRAPHY: A COMPARISON OF OPTICAL AND INFRARED VELOCITY DATA FOR THE MIRA VARIABLE CHI CYNGI

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High-dispersion optical velocities for χ Cyg obtained during the interval of 1959 to 1979 are presented. Particularly important are data for 1977, when Fourier transform spectra were obtained by Hinkle, Hall, and Ridgeway (1982) and for the faint maximum of 1966, when an unusual emission spectrum similar to that of R Cyg at faint maxima was found.

An analysis of the velocities of the various observed layers under the constraint that the temperature must decrease outward (except at a shock front where emission is produced by a thin hot layer) produces a consistent kinematic model with a shock first seen near phase 0.9 and disappearing by phase 0.6.

The emission spectrum of 1966 showed lines seen previously only during faint maxima of R Cyg. As in R Cyg, the emission lines during a faint maximum show a very small velocity of expansion which we attribute to enhanced limb brightening.

Key words: stellar shock waves–layering in stellar atmospheres–velocity analysis of atmospheric structure

I. Introduction

The spectra of long-period variables have presented a number of problems that have perplexed astrophysicists for many yaears (Merrill 1960). Recently great progress has been made by high-resolution observations in the near infrared which have resolved multiple components in absorption and allowed the identification of the main pulsating layers (Hinkle, Hall, and Ridgeway (1982), henceforth HHR; Hinkle Scharlach, and Hall (1984), henceforth HSH). At the same time, substantial progress has been made with the theory of shockwaves in the atmospheres of cool giants (Willson and Hill 1979; Hill and Willson 1979; Gillett and Lafon 1983, 1984; Bertschinger and Chevalier 1985; Fox and Wood 1985).

There remain considerable difficulties in understanding the kinematics of the various layers identified in χ Cygni by HHR. In addition to the main pulsating layers, they found lines from a circumstellar envelope and two additional layers whose kinematics are difficult to interpret. One layer shows a velocity of -8 km s^{-1} with a rotational temperature of 800 K, while the second shows a velocity near 0 and a rotational temperature of 1500 K, respectively.[1] While the existence of these two cool layers above the main shock might be interpreted as evidence for a second shock (Willson and Hill 1979; Hill and Willson 1979), their temperatures do not support that interpretation. As observed in the main shock, the gas that is falling in should have the cooler temperature, but the opposite is observed for the 800 K and 1500 K layers in χ Cyg. HHR interpret those layers using the reasonable assumption that the temperature decreases outward so that the 800 K layer is feeding both the circumstellar envelope and the 1500 K gas which is falling back into the star at about 7.5 km s^{-1}. The problem with this interpretation is that the 800 K layer is losing gas in both directions but has no evident supply of material.

As part of a planned cooperative effort with HHR, we obtained a number of spectrograms of χ Cyg during the 1977 observing season and have combined those data with other optical spectra, including plates obtained at Palomar by A. J. Deutsch between 1959 and 1966. A preliminary report was presented in 1978 where some details, not shown here, are to be found (Wallerstein 1979).

II. Observational Data

The primary purpose of this study is to identify the measured radial velocities in the optical region with the various layers isolated by HHR in the near infrared. Since the main criterion for identification is similarity of radial velocity, we have made extensive velocity measurements using the Grant measuring machines at the California Institute of Technology and the Joint Institute for Laboratory Astrophysics. The data are summarized in Table I, where we list the measured velocities as well as the date and equipment utilized to obtain each plate. The Palomar spectrograms of 1959–66 were obtained by A. J. Deutsch. Some of his 9 Å mm^{-1} spectrograms were extremely heavily exposed in order to reveal circumstellar cores; however, they usually showed asymmetries without genuine resolution. It appears that a dispersion of 6 Å mm^{-1} is necessary to resolve the cores. The spectrograms of 1970–71 have been described elsewhere (Wallerstein 1975) and are included here only for completeness. We concentrated here on the velocities of absorption lines

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[1]These velocities are recorded with respect to the sun. From thermal microwave emission the star's center-of-mass velocity is -7.5 km s^{-1}.

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TABLE I

Optical Velocity Data for Chi Cygni

Date	Days from maximum	Mag of max.	Disp./Region	Observatory	Radial Velocities (km s^{-1}) Emission Lines H v_r	±	n	Emission Lines metals v_r	±	n	Absorption Lines excited v_r	±	n	Absorption Lines resonance v_r	±	n	Notes
8-18-59	-26	4.9	9/B	P							+3	0.5	24				
8-25-59	-19	4.9	9/B	P										-10	±1	4	
10-01-60	-28	5.0	9/B	P				-7	±2	4	-5*	0.9	6	-9	±1	6	
10-04-60	-25	5.0	9/B	P							-4*	0.5	13	-9	±1	13	
06-11-65	+80	4.2	7/R	P							+1.5		2				
05-05-66	+3	6.0	7/R	P	-39		1	-11	±1	9	-1	1.4	5	-14		2	+
09-09-70	-51	5.2	18/R	KPNO							-8*						
09-12-70	-58	5.2	27/IR	KPNO							+7	2.5	4	-17		1	‡
10-09-70	-21	5.2	6/R	Mt.W.	-36		1				-4	0.6	7				§
11-23-71	-31	4.8	10/B	Mt.W.	-16	±4	3	-20		2	+0.5	0.8	15	-16		2	‖
11-24-71	-30	4.8	6/R	Mt.W.							-5*		1				
12-16-71	-8	4.8	9/B	P	-15	±2	6	-14		2	0.5	0.7	12	-15	±1.5	4	**
12-17-71	-7	4.8	13/Y	P	-19		1	-13		1				-16		2	†
03-24-75	-59	4.4	4/IR	KPNO										-16		2	‡
04-27-75	-25	4.4	6/B	DAO	-15		2	-15	±2	14	+1.7	0.9	13				
08-12-77	-12	4.9	2.4/B	DAO	-18	±7	3	-15	±2	19	+1.0	0.4	24	-13	±0.9	6	††
08-13-77	-11	4.9	5/Y	DAO										-17		2	†
08-13-77	-11	4.9	10/R	DAO	-32		1				0.0	0.4	33	-16		2	†·
08-14-77	-10	4.9	6/B	DAO	-14	±4	5	-10		2	-1.6	0.6	30	-14	±0.7	3	‖
08-31-77	+7	4.9	13/IR	P										-18		1	‡
09-01-77	+8	4.9	7/IR	P				-20		2	-0.5	0.7	15				
09-21-77	+27	4.9	6/B	DAO	-17	±3	6				-0.1	0.3	23	-12	±1.0	3	‖
10-05-79	-45	5.2	10/R	DAO							-6.2*		1				‡‡

* These entries are for relatively weak ground multiplet lines
† Resonance lines are NaI
‡ Resonance line is KI
§ TiO absorption shows -4.4 km s^{-1}
‖ Resonance lines are CrI
** Resonance lines are CrI and CaI
†† Resonance lines are MnI and CrI
‡‡ TiO absorption shows -9.9 km s^{-1}

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because emission velocities have been described extensively by P. W. Merrill (1947, 1953); however, the unusual emission spectrum seen during the faint maximum of 1966 is discussed in Section III. D.

III. Discussion

It is clear from Table I that we have seen at least three layers of gas. These are a circumstellar layer represented by strong resonance lines, the so-called photospheric layer seen in absorption lines arising from excited levels, and an emitting layer. In addition, we will discuss the negatively displaced absorption lines seen first by Maehara (1968). After describing some of the characteristics of each layer, we will endeavor to relate them to the infrared observations of HHR and to a reasonable kinematic model.

A. *The Circumstellar Gas*

If we take a mean of the velocities of the resonance lines, omitting relatively weak lines such as λ6572 of Ca I and partially resolved lines on 9Å mm^{-1} plates, we find $V_{CS} = -15.3 \pm 0.65$ from nine entries. This is exactly equal to the V_{CS} of HHR (Table 4). This may be compared with a value of -17.2 ± 0.9 from six observations of thermally excited molecular emission, omitting one observation of the rather weak SiO J = 3→2 line (HSH 1984). The small difference may be understood in terms of our measurement of the center of an absorption line, while the microwave value is determined by the extreme edge of a fitted line profile. Actually, the difference is significant and is clarified by the resolution by HHR of the circumstellar component into two shells of velocity −13.9 and −17.6 km s^{-1} on an interferogram with $\lambda/\Delta\lambda$ = 100,000.

Sanner (1977) reported sodium and potassium emission during and after the rather faint maximum ($m_v = 5.4$) of 1976. The observed P Cygni profiles showed emission centered at −6.3 km s^{-1} and absorption at −17.1 km s^{-1}.

The DAO and Palomar plates of August 1977 are of sufficient resolution and quality to provide accurate radial velocities for circumstellar lines of five different species. We list the data in Table II in order of ionization potential for the neutral species. A correlation is quite evident, indicating that the Mn I and Cr I lines are largely contributed by the shell at −13.9 km s^{-1}, while the lines of Na I and K I are to be associated with the shell at −17.6 km s^{-1}. This suggests that Na I and K I are substantially ionized in the former shell, which agrees with the presence of Sr II with nearly the velocity of that shell. None of the circumstellar features, except possibly the Sr II lines, appear to be associated with the 800 K gas, whose velocity is −8 km s^{-1}.

TABLE II
Velocities of Displaced Cores of Resonance Lines at Light Maximum of 1977

Element	Ionization Potential (eV)	Vr (kms^{-1})	±	n
Mn I	7.4	-12.3	1.1	3
Cr I	6.7	-14.6	0.6	3
Na I	5.1	-17.2	0.8	2
K I	4.3	-18.2	...	1
Sr II	11.0*	-11.6	0.6	2

*The ionization potential of Sr I is 5.7 eV.

From eight measurements of the centers of thermally emitting molecular gas quoted by HSH, the center of mass velocity of χ Cyg is -7.5 ± 0.4 km s^{-1}, indicating an expansion of the two envelopes of 6.4 and 10.1 km s^{-1}.

For several supergiants and semiregular variables Bernat (1981) found multiple circumstellar components with a variety of rotational temperatures between 60 K and 550 K. Apparently irregular or episodic expulsion of matter is common among a variety of types of luminous stars.

B. *The Photospheric Layer*

Radial velocities of lines from excited levels must represent a layer considerably deeper than the circumstellar envelope. If we take a mean of all plates on which at least ten lines were measured, we find $V_{abs} = -0.1 \pm 0.6$ km s^{-1}. This layer, so readily measurable in absorption in the blue-violet region of the spectra of long-period variables, has generally been called the photospheric layer, although its relationship to the pulsation process is not entirely clear.

By matchng velocities, this layer is most easily identified with the layer at $V = +0.1 \pm 0.9$ km s^{-1}, seen in the $\Delta V = 1$ CO lines by HHR (Table 4). The possible identification with the weakening CO features of steadily increasing velocity traced by HHR from phase 0.0 through a complete cycle to 0.10 is deemed to be unlikely by HHR because of the discrepancy in velocity of 8 km s^{-1} near phase 0.90 and the fact that the CO component in question disappears after phase 0.10 while the layer seen in the blue-violet region is observed continuously throughout the cycle (Merrill 1947, 1953). However, the model is simplified if we permit the identity of these two layers (see end of § III. E).

C. *The Layer Seen in the Near Infrared*

The third absorption system detectable in the optical region was discovered by Maehara (1968) soon after maximum light with a radial velocity near −20 km s^{-1} and traced with increasing velocity to phase 0.30. Its obvious analog in CO is the component of high excitation, first seen at the same phase and velocity by HHR (Fig. 12 and 16) and then traced through a little more than one complete cycle. Our best infrared spectrogram, Pb 14641 (date: 1977 September 1), showed only one component in absorption at − 0.5 km s^{-1}, which may be identified

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either with the photospheric velocity discussed above or with the end of the cycle that started during the previous maximum and was traced by HHR in the CO $\Delta C = 3$ bands. Since Rayleigh scattering causes the opacity to be lower at λ8000 than at λ4000, the layers seen by Maehara must be at least as deep, and probably deeper, than those providing the absorption lines in the λ4000 region.

D. *The Emission-Line Velocities*

Though this study was primarily oriented toward understanding the various absorbing layers in χ Cyg, we have also reported some emission-line velocities.

The emission spectrum of χ Cyg was observed extensively by Merrill (1947, 1953). It shows all the complex characteristics of LPV emission spectra, especially the effects of overlying absorption and selective excitation, both of which change with phase. However, the radial velocity of the emission lines remains almost constant near -15 km s^{-1} from the first appearance of the lines near phase 0.90 until phase 0.50.

From Table I we find a hydrogen emission velocity of -15.8 ± 0.7 by averaging only those six entries with at least two lines. For the metals we find $V = -15.6$ from the only three plates with more than four lines (omitting Pb 9374, taken on 1966 May 5). These values are very similar to numerous emission-line velocities obtained over many years for χ Cyg. We will now discuss the discrepant emission velocities.

The Hα emission line is discrepant because an unidentified molecular band absorbs the long wavelength part of the line. This phenomenon is also seen clearly in U Orionis (Wallerstein 1985) but not in the pure S star R Cyg (Wallerstein et al. 1985) so the band is probably TiO or possibly VO. It first appears at type M5 or M6. This distortion of Hα in long-period variables has been discussed by Joy (1947) and Bretz (1967).

The emission velocity of -7 km s^{-1} measured on Pc 5377 (taken on 1960 October 1) comes from the violet resonance lines of Al I and Ca II only. The apparent velocity is probably displaced to the red because of absorption by the circumstellar shell.

The emission spectrum during the very low maximum of 1966 is particularly interesting. The emission velocity of -11.2 ± 0.7 from nine lines on plate Pb 9374 (date: 1966 May 5) is well determined. It was observed within three days of one of the faintest maxima ever recorded for χ Cyg, visual magnitude 6.0. The lines seen in emission are the same as the strongest ones measured by Deutsch and Merrill (1959) in R Cyg and remeasured and rediscussed by Wallerstein et al. (1985). Our single velocity measurement might associate this layer with the 800 K layer whose velocity is near -8 km s^{-1} in χ Cyg. On the other hand, the extensive discussion of the same phenomenon during faint maxima of R Cyg indicated that the low values of expansion shown by the emission spectrum of R Cyg indicated that an enhancement of limb brightening of the emission lines may well be the reason for their approaching the center of mass velocity during faint maxima.

An alternative source for the emission spectrum during faint maxima is the 800 K gas which has the right velocity. The low rotational temperature of CO, which must indicate the local kinetic temperature, ensures that the emission lines are due to selective excitation by precursor radiation emerging from the shock front (Gillet and Lafon 1984).

Sodium and potassium emission were first seen in χ Cyg during the 1966 cycle by Bretz (1967, especially Figs. 2 and 3). She notes how rare such emission is in long-period variables. Potassium emission has also been seen in U Ori during its low maximum of 1983 (Wallerstein 1985). Sodium emission is evident on plate Pb 9374, being stronger on the violet than the red side and cut by a deep absorption component (as also described by Bretz on the basis of a lower-dispersion spectrogram).

E. *Stratigraphy of χ Cyg*

To understand the kinematics of the atmosphere of χ Cyg, we must locate the various layers recognized by their velocities and excitation temperatures. We can be assisted in this procedure by knowledge of the wavelength dependence of the continuous opacity, since the opacity determines how deeply into the atmosphere we may look. The opacity almost certainly reaches a minimum at 1.6 μm and increases toward the violet due to H^- and Rayleigh scattering, while it increases to longer wavelengths due to free-free transitions of H^-.

We have recognized three distinct absorbing layers in the optical (roughly λ = 4000 Å to 8500 Å) region and identified each with one of the layers observed in CO by HHR. One additional layer has been observed by HHR, but not by us. It has a very low rotational temperature, 800 K, and appears to be stationary with respect to the center of mass of the star. We have not recognized this layer in the optical spectrum, probably for two reasons. At 800 K grains are forming, thereby depleting the metals whose lines we have looked for. Secondly, from HHR the 800 K layer produces lines of about 20 km s^{-1}, FWHM. Such broad lines would be hard to recognize, especially on the highest-dispersion spectrograms.

The velocity structure of χ Cyg is shown in Figure 1, where we have plotted all the near infrared CO, optical and radio data in one diagram. The location of the various layers in order of height depends on the adopted model. HHR place the various layers in order of increasing temperature, as shown in the left half of Figure 2.

This interpretation faces two problems. According to HHR, the 800 K layer which is stationary with respect to the center of mass of the star feeds material upward into the circumstellar envelope and downward via the 1500 K

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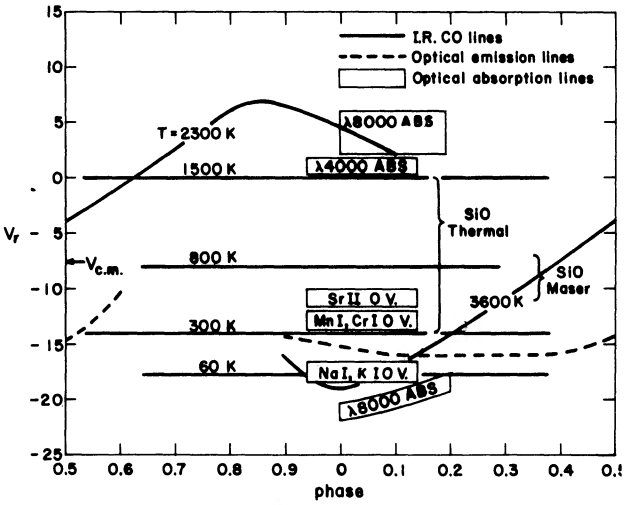

FIG. 1–The velocity structure of χ Cyg during one cycle. The various components of CO (HHR), optical absorption (Maehara 1968; Pierce, Willson, and Beavers 1979; this paper), optical emission (this paper; Merrill 1947, 1953), and molecular emission (Snyder and Buhl 1975; Dickinson et al. 1978; Knapp et al. 1982) are shown. The velocities are not corrected for any projection or limb effects.

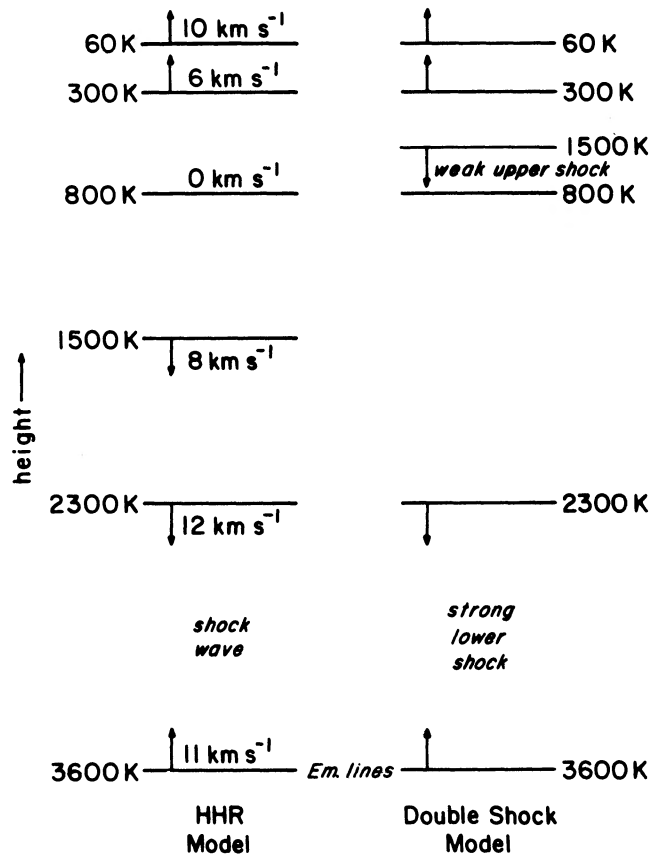

FIG. 2–Two possible stratigraphies for χ Cyg near maximum light. The figure shows the ordering of the layers and their direction of motion, but not their separation or thickness. The layering on the left is that suggested by HHR, while on the right we show a two-shock model along the lines of Willson and Hill (1979) and Hill and Willson (1979).

layer. There is no obvious source of material for this layer, nor is a means of support suggested to keep it suspended above the pulsating layers over at least the two cycles covered by HHR. Radiation pressure on grains forming at 800K would be the most likely possibility. The second problem is the lack of optical absorption lines from this layer when the 1500K layer below it is easily seen at all wavelengths, especially in the blue-violet region. This argument is not entirely decisive, however, because the radial velocity of the Ca I zero-volt line at λ6572 is often near -8 km s^{-1} and two measures of rotational lines of TiO in the red region yielded -4.4 and -9.9 km s^{-1}. The Ca I velocities can be explained as a blend of photospheric and circumstellar gas and one plate, Pc 8731 (1965 June 11), Ca I (and Sc I λ6307) showed velocities at the photospheric value.

The alternative interpretation of the relationship of the 800K and 1500K layers is that the 1500K layer is the higher one and that they represent the layers separated by a weak upper shock, as suggested by Willson and Hill (1979) and Hill and Willson (1979). We show this on the right side of Figure 2. This explanation fails to account for the fact that the upper, infalling layer has the higher temperature, while shock theory always predicts that the lower layer should be heated by the shock. This interpretation reduces the problem of the lack of absorption lines from the 800K layer in the blue region, but does not entirely eliminate the problem, since that layer was not recognized by Maehara in the near infrared. Perhaps it was not present during Maehara's observations. The presence of Ca I and TiO with suitable velocities may alleviate the problem in this model as well as it does in the HHR model.

The velocity of the "photospheric" layer in 1944 decreased gradually to near -6 km s^{-1} by phase + 162 days (Merrill 1947). In 1951 the absorption velocity was again near -5.5 during the interval of 76 to 32 days before minimum, but had returned to zero by minimum light (Merrill 1953). It is not clear how these velocity changes may be related to the component at -8 km s^{-1} and $T = 800$K of HHR, but it showed an outward acceleration (actually, a retardation of its rate of fall) of the "photospheric" layer during the cycle. For an upper-shock interpretation the positive velocity (relative to the centroid of the star) predicts that this is the *upper* layer which should be accelerated by gravity, i.e., inward.

While the HHR model for χ Cyg seems to handle the situation rather well at maximum light (except for the origin of the 800K layer), an extension of it during declining light runs into a serious problem. The main CO absorbing layer is decelerated from its velocity of -19 km s^{-1} at maximum light through phase 0.4, when it equals the center-of-mass velocity and begins to fall backward; the optical emission velocity is not decelerating, but rather, remains near -16 km s^{-1} until it rises to a point near the center-of-mass velocity between phases 0.5 and

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0.6. Considering the uncertainties of measurement and the difference between the projection and limb-darkening effects for emission and absorption lines, it is reasonable to say that the shock responsible for the emission lines is at the upper edge of the $T = 3600$K CO layer near maximum light. However, after phase 0.15 the emitting material begins to run ahead of the absorbing layer and continues to do so to an increasing extent, as the absorbing layer falls back toward the star after phase 0.4. The turnup of the emission velocity between phases 0.5 and 0.6 may not be a real velocity change, but rather a manifestation of increasing limb brightening as the emitting layer rises higher in the atmosphere.

For the shock wave, whose flow velocity is given by the emission lines, to remain luminous, it must be sweeping up infalling gas. The only layer available for this after phases 0.1 is the 1500K layer. In fact, the similarity in velocity and temperatures of the 1500K layer and the 2300K layer suggests that they may be the same gas. This means that the 1500K layer is fed by the 2300K layer rather than the stationary 800K layer. It seems more likely that the gas would cool off during the cycle than that it would heat up after falling from the mysterious 800K "reservoir" that is hypothesized to be floating above the pulsating and emitting shocked gas. The 800K gas would then be supplied by the dying shock whose emission weakens as its velocity decreases (assuming that part of the change in velocity between phases 0.5 and 0.6 is real).

We show this sequence of events in Figure 3. The only discrepancy of this model with the observations is the simultaneous existence of the 1500K layer and the 2300K layers around phases 0.8 to 0.9. Our model predicts that the 1500K layer should be disappearing as the emission spectrum and hence the shock dies out around phase 0.6. At this phase the radial velocities of the 1500K and 2300K layers are the same and hence the existence of the 1500 K layer cannot be confirmed. However, HHR (Table 2) interpret their profiles to indicate that near phase 0.9 the layers with velocities near +6 km s^{-1} and 0 km s^{-1} are distinct, though they note some uncertainty by the use of colons after the velocities quoted for $v = 3 \rightarrow 1$ moderate excitation lines in their Table 2. If their resolution into separate components is not correct, this model would be fully consistent with the phenomena.

We do not have spectrograms of the highest resolution at phase 0.9, when the separation reaches 6 km s^{-1}. Our highest-resolution spectrogram was obtained at phase 0.97, when the separation was about 4 km s^{-1}, which is a little less than the resolution of the 2.4Å mm^{-1} plate on which the absorption lines are somewhat intrinsically broadened.

F. *Some Calculated Distances*

It is of interest to note that an integration of the velocity curve of the $\Delta v = 3$ CO lines yields an expansion of 0.83 AU between phase 0.9 and maximum expansion at phase 0.40. The actual expansion must be greater, since no projection effect has been used. The reality of such distances may be questioned because the same gas is not seen throughout the cycle, but rather, material is flowing through $\tau_{\rm line} = 1$ due to the pulsation.

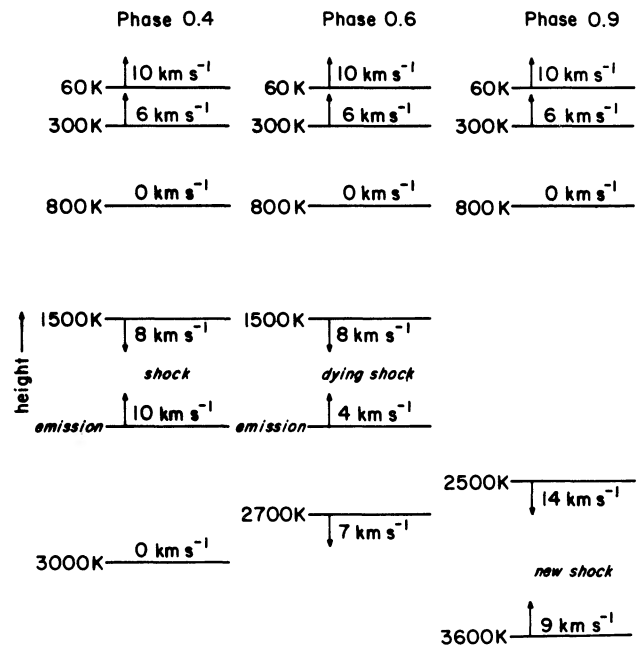

FIG. 3–Suggested stratigraphy for χ Cyg during phases 0.4, 0.6, and 0.9.

For the emission velocity, an integration from first appearance at phase 0.9 to disappearance at phase 0.6 yields a distance of 1.2 AU. The necessity of including limb brightening for optically thin emission lines indicates that a somewhat larger distance is called for. However, the emission lines do not come from the same gas throughout the cycle, but rather, are excited at progressively higher layers by the emerging shock. The shock's velocity is unobservable, but is probably a few km s^{-1} faster than that of the emitting gas. If it is 5 km s^{-1} faster than that of the emitting gas, the shock travels at least 2 AU between phases 0.9 and 0.6.

IV. Summary

We have shown that it is possible to identify and understand most of the layers seen in absorption in χ Cyg in both the optical and infrared regions, provided that the 1500K and 2300K layers of HHR may be identified as actually being the same gas. However, the 800K level of HHR is the most difficult one to understand. Its origin and especially its support at very nearly the systemic velocity over several cycles remain unclear.

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